CHAPTER 7

ALTERNATING CURRENT

CHAPTER ESSENCE AT A GLANCE:
Sinusoidal AC voltage and current phasors, series LCR resonance, power factor, and efficient voltage transformation using step-up/step-down transformers.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Root Mean Square (RMS) Values:** Steady direct current that produces the identical average heating effect in a given resistor over a complete alternating cycle.

• **Phase Relationships & Reactance:** In a purely resistive circuit current and voltage remain in phase; in an inductive circuit voltage leads current by ninety degrees, whereas in a capacitive circuit current leads voltage by ninety degrees.

• **Series Resonance & Minimum Impedance:** Resonance occurs when inductive and capacitive reactances cancel, dropping total impedance to pure resistance and maximizing current.

• **AC Power Factor & Wattless Current:** Average power dissipation depends on the phase angle between voltage and current; at ninety degrees phase difference, zero average power is dissipated.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}, \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

RMS values of alternating current and voltage

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}$$

Inductive and capacitive reactances

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Impedance of series LCR circuit

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Resonant frequency of series LCR circuit

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Current in an ideal inductor or capacitor consumes ZERO average power because phase difference ϕ = 90° cos 90° = 0.

• **Alert:** [Valuation Rubric] Draw clear phasor diagram showing V R along current, V L leading by 90°, V C lagging by 90° .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Relation between peak value and RMS value of AC current/voltage [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[RMS Definition]** Root Mean Square (RMS or virtual/effective) value of AC is that steady DC current which produces the same quantity of Joule heat in a given resistance in a given time as is produced by the AC in one complete cycle. [1 Mark]

• **[Derivation of I rms = I_0 divided by sqrt(2)]** Let alternating current be $i = I_0 \sin(\omega t)$. Heat generated in resistance R in time dt: $dH = i^2 R\, dt = I_0^2 R \sin^2(\omega t)\, dt$. Total heat in one time period T: $H = \int_0^T I_0^2 R \sin^2(\omega t)\, dt = I_0^2 R \int_0^T [(1 - \cos 2\omega t) \text{ divided by } 2]\, dt = (I_0^2 R \text{ divided by } 2) [t - \sin(2\omega t) \text{ divided by } (2\omega)]_0^T = (I_0^2 R \text{ divided by } 2) [T - 0] = (\text{half } (0.5)) I_0^2 R T$. By definition, $H = I_{rms}^2 R T$. $I_{rms}^2 R T = (\text{half } (0.5)) I_0^2 R T$ implies I rms = I_0 divided by sqrt(2) approx. 0.707 I_0. Similarly, V rms = V_0 divided by sqrt(2) approx. 0.707 V_0. [2 Marks]

• **Key Exam Result:** I rms = I_0 divided by sqrt(2) = 0.707 I_0; V rms = V_0 divided by sqrt(2) = 0.707 V_0

2. AC circuit containing pure resistor: phase relation and phasor diagram [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Circuit & Equations]** Pure resistor R connected across $v = V_0 \sin(\omega t)$. Instantaneous current: i = v divided by R = (V_0 divided by R) $\sin(\omega t) = I_0 \sin(\omega t)$, where $I_0 = V_0$ divided by R. [1 Mark]

• **[Phase Relation & Phasor Diagram]** Voltage and current are IN PHASE (phase difference $\Phi = 0$). Phasor diagram: Phasor V and phasor I lie along the exact same line, rotating at angular speed ω. Average power dissipated: P avg = V rms I rms $\cos(0°)$ = V rms I rms = I rms^2 R. [1 Mark]

• **Key Exam Result:** Pure Resistor: v and i are in phase ($\Phi = 0$); P avg = V rms I rms

3. AC circuit containing pure inductor: inductive reactance X L = \ω L and phase [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Circuit & Derivation]** Pure inductor L connected across $v = V_0 \sin(\omega t)$. By Kirchhoff's loop rule: v - L (di divided by dt) = 0 implies di = (v divided by L) dt = (V_0 divided by L) $\sin(\omega t)\, dt$. Integrating: i = (V_0 divided by L) $\int \sin(\omega t)\, dt$ = - (V_0 divided by ω L) $\cos(\omega t)$ = (V_0 divided by ω L) $\sin(\omega t - \pi$ divided by 2). Peak current: $I_0 = V_0$ divided by X L, where Inductive Reactance $X L = \omega L = 2\pi f L$ (Unit: Ω). [2 Marks]

• **[Phase & Phasor]** Current LAGS behind voltage by phase angle π divided by 2 (90°). Phasor I is 90° clockwise behind phasor V. Average power P avg = V rms I rms $\cos(90°) = 0$. [1 Mark]

• **Key Exam Result:** $X L = \omega L = 2\pi f L$; Current lags voltage by 90° (π divided by 2 rad); P avg = 0

4. AC circuit containing pure capacitor: capacitive reactance X C = 1 divided by (\ω C) and phase [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Circuit & Derivation]** Pure capacitor C connected across $v = V_0 \sin(\omega t)$. Instantaneous charge: $q = C v = C V_0 \sin(\omega t)$. Current i = dq divided by dt = d divided by dt [$C V_0 \sin(\omega t)$] = $\omega C V_0 \cos(\omega t)$ = (V_0 divided by [1 divided by (ω C)]) $\sin(\omega t + \pi$ divided by 2). Peak current: $I_0 = V_0$ divided by X C, where Capacitive Reactance X C = 1 divided by (ω C) = 1 divided by ($2\pi f C$) (Unit: Ω). [2 Marks]

• **[Phase & Phasor]** Current LEADS voltage by phase angle π divided by 2 (90°). Phasor I is 90° counter-clockwise ahead of phasor V. Average power P avg = V rms I rms $\cos(90°) = 0$. [1 Mark]

• **Key Exam Result:** X C = 1 divided by (ω C) = 1 divided by ($2\pi f C$); Current leads voltage by 90°; P avg = 0

5. Why is high voltage AC preferred for long distance power transmission? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Reason for High Voltage AC Transmission]** Electrical power P transmitted over transmission line of resistance R at voltage V: Current carried by cables: I = P divided by V. Power lost as heat in transmission cables: P loss = $I^2 R$ = (P divided by V)^2 R = (P^2 R) divided by V^2. Since transmission power loss is inversely proportional to V^2 (P loss proportional to 1 divided by V^2), stepping up voltage by a factor of 10 reduces transmission power loss by a factor of 100! Furthermore, lower current allows using much thinner conductors, drastically reducing copper infrastructure costs. [2 Marks]

• **Key Exam Result:** P loss proportional to 1 divided by V^2; High voltage drastically cuts Joule heating loss and cable weight

6. Numerical on LCR impedance, current, and phase angle [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Circuit & Reactances]** Series LCR with R = 30 Ω, L = 0.4 H, C = 100 μF = 10^{-4} F connected to 200 V, 50 Hz AC. Angular frequency: ω = 2π f = 2 × 3.1416 × 50 = 314.16 rad/s. X L = ω L = 314.16 × 0.4 = 125.7 Ω. X C = 1 divided by (ω C) = 1 divided by (314.16 × 10^{-4}) = 10000 divided by 314.16 = 31.8 Ω. [1.5 Marks]

• **[Step 2: Impedance, Current, and Phase]** Net reactance: X = X L - X C = 125.7 - 31.8 = 93.9 Ω. Impedance: Z = sqrt(R^2 + X^2) = sqrt(30^2 + 93.9^2) = sqrt(900 + 8817) = sqrt(9717) = 98.6 Ω. RMS Current: I rms = V rms divided by Z = 200 divided by 98.6 = 2.03 A. Phase angle: tan Φ = (X L - X C) divided by R = 93.9 divided by 30 = 3.13 implies Φ = 72.3° (current lags voltage). [1.5 Marks]

• **Key Exam Result:** Z = 98.6 Ω; I rms = 2.03 A; Φ = 72.3° (lagging)

7. Numerical on resonant frequency and Q-factor [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Data & Formulas]** L = 2.0 H, C = 32 μF = 32 × 10^{-6} F, R = 10 Ω. Resonant angular frequency: ω_0 = 1 divided by √(L C) = 1 divided by √[2.0 × 32 × 10^{-6}] = 1 divided by √[64 × 10^{-6}] = 1 divided by (8 × 10^{-3}) = 125 rad/s. Linear resonant frequency: f_0 = ω_0 divided by (2π) = 125 divided by (2 × 3.1416) = 19.9 Hz. [1.5 Marks]

• **[Step 2: Q-Factor Calculation]** Q = (ω_0 L) divided by R = (125 × 2.0) divided by 10 = 250 divided by 10 = 25. Or via: Q = (1 divided by R) sqrt(L divided by C) = (1 divided by 10) sqrt(two-thirds2×10^{-6}) = (1 divided by 10) sqrt(62500) = 250 divided by 10 = 25. [1.5 Marks]

• **Key Exam Result:** Resonant frequency f_0 = 19.9 Hz; Q-factor = 25